

THE NEUMANN PÓLYA INEQUALITY ON SIMPLY CONNECTED PLANAR DOMAINS

1. INTRODUCTION

Let $\Omega \subset \mathbb{R}^2$ be a bounded connected Lipschitz domain, and let $|\Omega|$ denote its area. The Neumann Laplacian $A_N = -\Delta_\Omega^N$ is the self-adjoint operator associated with the quadratic form

$$q[u] = \int_\Omega |\nabla u|^2, \quad u \in H^1(\Omega).$$

Its eigenvalues, repeated according to multiplicity, are denoted by

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \leq \mu_2(\Omega) \leq \cdots.$$

We include the zero eigenvalue in the strict counting function

$$N_N(E) = \#\{j \geq 0 : \mu_j(\Omega) < E\}.$$

Weyl's law identifies the leading term as $N_N(E) \sim |\Omega|E/(4\pi)$ as $E \rightarrow \infty$; see [26, 5]. Pólya's conjecture asks whether this asymptotic expression is a lower bound at every positive energy.

The conjecture goes back to Pólya's discussion of vibrating membranes in [18]. His subsequent paper [19] established the corresponding inequalities for tiling domains, with an additional restriction in the Neumann case that was removed by Kellner [11]. In the indexing used here, the planar Neumann part of the conjecture takes the following form.

Conjecture 1. *For every bounded connected Lipschitz domain $\Omega \subset \mathbb{R}^2$ and every $j \geq 1$,*

$$|\Omega|\mu_j(\Omega) \leq 4\pi j.$$

The same problem can be stated as $N_N(E) \geq |\Omega|E/(4\pi)$ for all $E > 0$. The strict version, in which the eigenvalue inequality is replaced by $|\Omega|\mu_j(\Omega) < 4\pi j$, is called the strong Pólya conjecture in [8]. The question concerns the whole spectrum; an asymptotic formula alone does not settle the low- and intermediate-energy ranges.

Sharp geometric estimates for the first positive Neumann eigenvalue were obtained by Szegő [23] for simply connected planar domains and by Weinberger [25] without that topological restriction and in arbitrary dimension. For the full spectrum, Kröger [12] proved sharp bounds for eigenvalue sums and upper bounds for individual eigenvalues with a larger constant. Laptev [13] developed the associated trace inequalities and product-domain arguments. These results do not give the conjectured constant for individual eigenvalues on an arbitrary planar domain.

Beyond tiling domains, Filonov, Levitin, Polterovich, and Sher [5] proved the conjecture for the disk and for arbitrary planar sectors, as well as its Dirichlet version for balls in all dimensions. Freitas and Salavessa [9] obtained further families of non-tiling domains by using thin-domain constructions. He and Wang [10] proved the conjecture for sufficiently thin product domains. More recently, Filonov [4] improved the universal Neumann counting bound for planar convex domains. Frank and Larson [7] extended sharp semiclassical inequalities on convex domains to a range of Riesz exponents below one. Here a Riesz mean of positive order α is $\sum_{j \geq 0} (E - \mu_j)_+^\alpha$, where $t_+ = \max\{t, 0\}$; bounds for such means are distinct from the order-zero counting inequality. The higher-dimensional Neumann ball case was established in [6]; see also [14]. None of these results covers all simply connected planar Lipschitz domains.

We prove the strong Neumann conjecture for this class.

Theorem 1. *For every bounded simply connected Lipschitz domain $\Omega \subset \mathbb{R}^2$ and every $j \geq 1$,*

$$(1.1) \quad |\Omega| \mu_j(\Omega) < 4\pi j.$$

Theorem 1 establishes Conjecture 1, with strict inequality, under the additional assumption of simple connectedness. Equivalently,

$$N_N(E) > \frac{E|\Omega|}{4\pi} \quad (E > 0).$$

Indeed, the counting inequality at $E = \mu_j$ gives (1.1), because $N_N(\mu_j) \leq j$. Conversely, if $m = N_N(E)$, then $m \geq 1$ and $E \leq \mu_m$, so (1.1) gives $E|\Omega| < 4\pi m$.

1.1. Formal verification. Theorem 1 and its equivalent strict counting formulation have been formalized and machine-checked in Lean 4.28.0. The development is available at <https://github.com/QuanyuTang/polya-neumann-simply-connected-formalization>. It identifies $\mathbb{R}^2$ with $\mathbb{C}$, defines the Neumann eigenvalues by the variational min–max construction, and proves the stated inequalities for bounded simply connected Lipschitz domains. The repository keeps the proof modular and provides a pinned Lean/Mathlib build together with an axiom audit for the main theorem and its principal intermediate results. The audited declarations use only Lean’s standard `propxt`, `Classical.choice`, and `Quot.sound` axioms; no `sorry` or `native_decide` proof step is used.

The proof uses a unitary evolution along the boundary. Its monodromy, defined in (2.1), is denoted by V_E , and we set $U_E = V_E^*$. For a smooth Jordan domain, $U_E - I$ is trace class and its Fredholm determinant is exactly $\exp(iE|\Omega|/2)$. The Neumann-to-Dirichlet map relates a Cayley transform of V_E to a boundary quadratic form. In conformal coordinates, the principal part of that form vanishes on the nonpositive Fourier modes. A transmutation of antiholomorphic functions, based on the method of Vekua [24, 17], identifies the corresponding kernel exactly. After a finite-rank correction at the cut in the boundary coordinate, this gives a uniform one-sided estimate for the Cayley transform. At a Neumann eigenvalue of multiplicity m , at most m dimensions are excluded from that estimate.

To describe the counting step, call an energy *nonresonant* if it is not a Neumann eigenvalue. At such an energy let $\Phi_-(E)$ be the sum, with multiplicities, of the arguments in $(-2\pi, 0)$ of the eigenvalues of U_E other than 1. Absolute convergence is proved in Section 7. The determinant identity makes

$$(1.2) \quad M(E) = \frac{E|\Omega|/2 - \Phi_-(E)}{2\pi}$$

an integer. The one-sided estimate shows that M is constant between Neumann eigenvalues and that its increase across an eigenvalue is at most its multiplicity. A separate small-energy calculation gives $M = 1$ below the first positive Neumann eigenvalue. The comparison $M(E) \leq N_N(E)$ then yields

$$N_N(E) \geq \frac{E|\Omega|}{4\pi} + \frac{\|U_E - I\|}{2\pi}.$$

The extra term is positive. Its continuity under boundary approximation is what preserves strictness for Lipschitz domains. The use of eigenphases has a precedent in inside–outside duality [3], but the boundary evolution here is not identified with an exterior scattering matrix. In particular, the argument does not require continuation of individual eigenvectors through the essential spectral point 1.

Section 2 constructs the evolution and establishes its relation to Helmholtz Cauchy data. Section 3 proves the transmutation and kernel results. Section 4 treats the periodic boundary form, and Section 5 proves the uniform estimate at nonresonant and resonant energies. Sections 6 and 7 give the normalization and the counting argument. Finally, Section 8 passes to Lipschitz boundaries using the smooth approximations of Ball and Zarnescu [1].

2. BOUNDARY TRANSPORT AND CAUCHY DATA

All Hilbert spaces are complex, and inner products are conjugate-linear in the first variable. We identify $\mathbb{R}^2$ with $\mathbb{C}$. For a bounded operator, an unqualified norm is its operator norm. Boundary coordinate measure is $d\theta$ on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$; direction measure is $d\phi/(2\pi)$. Sobolev duality extends the L^2 pairing. On boundary Fourier modes $\varphi_n = (2\pi)^{-1/2}e^{in\theta}$ put

$$D\varphi_n = n\varphi_n, \quad \Lambda\varphi_n = (1 + |n|)\varphi_n, \quad \mathcal{H}_+ = \overline{\text{span}}\{\varphi_n : n > 0\}, \quad \mathcal{H}_{\leq 0} = \mathcal{H}_+^\perp.$$

Set $\mathcal{H}_- = \overline{\text{span}}\{\varphi_n : n < 0\}$. The orthogonal projections onto $\mathcal{H}_+$, $\mathcal{H}_-$, and $\mathbb{C}\varphi_0$ are Π_+ , Π_- , and Π_0 , respectively; $\Pi_{\leq 0} = \Pi_- + \Pi_0$ and $D_+ = D|_{\mathcal{H}_+}$. We use $\|g\|_{H^s} = \|\Lambda^s g\|_{L^2}$ and set $X = H^{-1/2}(\mathbb{T})$, $X^* = H^{1/2}(\mathbb{T})$. The symbols $\mathcal{S}_1$ and $\mathcal{S}_2$ denote the trace-class and Hilbert–Schmidt ideals; $\det_F$ denotes the Fredholm determinant of an identity plus a trace-class operator.

We set $D^{-1}\varphi_0 = |D|^{-1}\varphi_0 = 0$ when these inverses occur. The spectrum and point spectrum of an operator are denoted by σ and σ_p , respectively. For a self-adjoint operator H , $\mathbf{1}_I(H)$ is its spectral projection on a Borel set I . The negative index $\text{ind}_- H$ is the supremum of the dimensions of subspaces on which its quadratic form is strictly negative. The boundary trace is denoted by Tr . We write tr for the operator trace. All estimates that are locally uniform in energy are for the fixed smooth domain under consideration.

For now Ω is a bounded C^∞ Jordan domain. Choose a positive smooth regular boundary parametrization γ on $[0, L]$, where $L = 2\pi$ is the parameter period, not the boundary length, and translate so that $\gamma(0) = \gamma(L) = 0$. On $\mathcal{H} = \ell^2(\mathbb{N}_0)$, with standard orthonormal basis $(e_n)_{n \geq 0}$, let

$$S_N e_0 = \sqrt{2}e_1, \quad S_N e_n = e_{n+1} \quad (n \geq 1), \quad [S_N^*, S_N] = 2P_0 - P_1 =: D_N.$$

Here $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ and P_n is projection onto $\mathbb{C}e_n$. For $E = k^2 > 0$ solve

$$(2.1) \quad W'_E = -\frac{ik}{2}(\gamma' S_N^* + \overline{\gamma'} S_N)W_E, \quad W_E(0) = I, \quad V_E = W_E(L), \quad U_E = V_E^*.$$

The coefficient in (2.1) is bounded and skew-adjoint, so $W_E(\theta)$ is unitary. At $E = 0$ set $W_0(\theta) = V_0 = U_0 = I$. We abbreviate the conormal density by $\partial_\nu^\gamma u = |\gamma'|(\partial_\nu u) \circ \gamma$, where ν is the outward unit normal. Define

$$(\mathcal{O}_E v)(\theta) = \langle e_0, W_E(\theta)v \rangle, \\ (\mathcal{T}_E g)(\theta) = \int_0^\theta \langle e_0, W_E(s)W_E(s)^* e_0 \rangle g(s) ds.$$

Initially $\mathcal{O}_E : \mathcal{H} \rightarrow L^2(0, L)$ and $\mathcal{T}_E : L^2(0, L) \rightarrow H^1(0, L)$. Their values are functions on the cut interval, not necessarily periodic. Directly,

$$(2.2) \quad \mathcal{O}_E^* g = \int_0^L W_E(s)^* e_0 g(s) ds, \quad \mathcal{T}_E + \mathcal{T}_E^* = \mathcal{O}_E \mathcal{O}_E^*, \quad \|\mathcal{O}_E\|_{\mathcal{S}_2}^2 = L.$$

Lemma 2.1. *The path $U_E - I$ is trace-norm continuously differentiable for $E \geq 0$ and trace-norm analytic for $E > 0$. If $L_E = -iU_E^* \partial_E U_E$, then*

$$(2.3) \quad \text{tr } L_E = |\Omega|/2, \quad \det_F U_E = \exp(iE|\Omega|/2), \quad L_0 = \frac{1}{2}|\Omega|D_N.$$

Proof. Set $G_x = -i(S_N + S_N^*)/2$ and $G_y = (S_N^* - S_N)/2$; then $[G_x, G_y] = iD_N/2$. Writing $\gamma = x + iy$, use the two coefficients $C_\theta = \sqrt{E}(x_\theta G_x + y_\theta G_y)$ and $\mathcal{C}_E = (xG_x + yG_y)/(2\sqrt{E})$. Differentiation of the evolution gives

$$\partial_\theta(W_E^* \partial_E W_E - W_E^* \mathcal{C}_E W_E) = W_E^* (\partial_E C_\theta - \partial_\theta \mathcal{C}_E + [C_\theta, \mathcal{C}_E]) W_E.$$

The mixed derivatives cancel, the endpoint values of $\mathcal{C}_E$ vanish, and $[C_\theta, \mathcal{C}_E] = -ibD_N/4$, where $b = \text{Im}(\overline{\gamma}\gamma')$. Hence

$$(2.4) \quad V_E^* \partial_E V_E = -iQ_E, \quad Q_E = \frac{1}{4} \int_0^L b(\theta) W_E(\theta)^* D_N W_E(\theta) d\theta, \quad L_E = V_E Q_E V_E^*.$$

The integral is a trace-norm integral of finite-rank operators and $\|Q_E\|_{\mathcal{S}_1} \leq 3 \int |b|/4$. Since $\int b = 2|\Omega|$ and $\text{tr } D_N = 1$, it has the asserted trace and limit at zero. The Picard series gives analyticity for positive energies and continuity at zero. Integrate $\partial_E V_E = -iV_E Q_E$ in trace norm from zero. The logarithmic derivative of the Fredholm determinant is $i \text{tr } L_E$, which proves (2.3); see also the standard determinant calculus in [22]. No sign of b has been assumed. $\square$

For direction density $a \in L^2(\mathbb{T}, d\phi/(2\pi))$ write

$$u_a(z) = \int_0^{2\pi} a(\phi) e^{-ik \text{Re}(ze^{-i\phi})} \frac{d\phi}{2\pi}, \quad h_a = u_a \circ \gamma, \quad g_a = |\gamma'| \partial_\nu u_a \circ \gamma.$$

Put $\partial = (\partial_x - i\partial_y)/2$ and $\bar{\partial} = (\partial_x + i\partial_y)/2$. With $y_0 = u_a/\sqrt{2}$ and $y_n = (2i/k)^n \partial^n u_a$ for $n \geq 1$, Parseval gives a vector in $\mathcal{H}$ with legitimate Hilbert-space derivatives. Since the outward normal is $-i\gamma'/|\gamma'|$,

$$(2.5) \quad y' = C_\theta y - \frac{i}{\sqrt{2}} g_a e_0, \quad h'_a = -ig_a - ik\gamma' y_1.$$

If $v = y(0)$, variation of constants and periodicity give

$$(2.6) \quad h_a = \sqrt{2} \mathcal{O}_E v - i \mathcal{T}_E g_a,$$

$$(2.7) \quad (I - V_E)v = -\frac{i}{\sqrt{2}} V_E \mathcal{O}_E^* g_a, \quad \mathcal{O}_E^* g_a = \sqrt{2} i (V_E^* - I)v.$$

Every $v \in \mathcal{H}$ occurs: if $a(\phi) = \sum_{n \in \mathbb{Z}} a_n e^{in\phi}$, take $a_0 = \sqrt{2}v_0$, $a_n = v_n$ for $n \geq 1$, and $a_n = 0$ for $n < 0$. These coefficients belong to $\ell^2(\mathbb{Z})$, so they define an L^2 density.

We use the layer-potential mapping and jump relations, Rellich's exterior uniqueness theorem, and uniqueness for zero interior Cauchy data; see [16]. Approximation by Herglotz waves is treated in [2]. The boundary-density statement needed here includes resonant energies and the $H^{-1/2}$ topology, so we give its layer-potential proof. We use the convention in which the interior trace minus the exterior trace of a double-layer potential with density h is $-h$, whereas the corresponding jump of the normal derivative of a single-layer potential with density g is g .

Lemma 2.2. *Fix $E > 0$. If $h \in H^1(\mathbb{T})$ and $g \in L^2(\mathbb{T})$ satisfy*

$$(2.8) \quad \langle g, h_a \rangle - \langle h, g_a \rangle = 0 \quad \text{for every } a,$$

then they are the Cauchy data of a unique interior Helmholtz solution $u \in H^{3/2}(\Omega)$. Also $\mathcal{O}_E$ is injective and

$$(2.9) \quad \dim \ker(V_E - I) \leq \dim \ker(-\Delta_\Omega^N - E).$$

If $\mu > 0$ has Neumann multiplicity m and $h_1, \dots, h_m$ are the boundary traces of an orthonormal basis $\phi_1, \dots, \phi_m$ of that eigenspace, the Herglotz normal traces at μ are dense in

$$(2.10) \quad X_0 = \{g \in X : \langle h_j, g \rangle = 0 \ (1 \leq j \leq m)\}.$$

At a nonresonant energy they are dense in X .

Proof. The outgoing single-layer potential of density $g/|\gamma'|$ minus the double-layer potential of density h has zero far field by (2.8), after conjugation and reversal of the plane-wave direction. It vanishes in the connected exterior. The jump relations give an interior H^1 solution with the prescribed data. Smooth-boundary regularity upgrades it to $H^{3/2}$. Two such solutions have zero Cauchy data and hence agree.

If $f = W_E v$ and $f_0 = 0$, its first row gives $0 = f'_0 = -(ik/\sqrt{2})\gamma' f_1$, and successive rows give every $f_n = 0$. This proves injectivity of $\mathcal{O}_E$. For $V_E v = v$, pair f with a driven Herglotz vector and integrate. One gets $\langle f_0, g_a \rangle = 0$ for every a . The negative of the double-layer potential of f_0 is zero outside and has interior trace f_0 and homogeneous Neumann data. This maps the fixed space injectively into the Neumann eigenspace and proves (2.9).

For density in X , an annihilator is a trace $h \in X^* = H^{1/2}$. The same double-layer argument identifies it with the trace of a Neumann eigenfunction. Conversely each such trace annihilates every g_a by Green's identity. The Hahn–Banach theorem proves the exact closure (2.10). The h_j are independent by zero-Cauchy-data uniqueness. This proof uses the dual topology $H^{-1/2}$, not merely an L^2 approximation assertion. $\square$

At a Neumann-nonresonant energy let $\mathcal{N}(E) : X \rightarrow X^*$ be the map $g \mapsto u \circ \gamma$ for $(-\Delta - E)u = 0$ and $|\gamma'| \partial_\nu u \circ \gamma = g$. By (2.9) the Cayley transform

$$\mathcal{K}_E = i(I + V_E)(I - V_E)^{-1}, \quad \text{Dom } \mathcal{K}_E = \text{ran}(I - V_E)$$

is densely defined and self-adjoint. Its resolvents are $(\mathcal{K}_E + i)^{-1} = (I - V_E)/(2i)$ and $(\mathcal{K}_E - i)^{-1} = (I - V_E)V_E^*/(2i)$, and are compact. Set $\mathcal{A}_E = 2\mathcal{N}(E) + i(\mathcal{T}_E - \mathcal{T}_E^*)$ on L^2 . Endpoint algebra in (2.7) gives

$$(2.11) \quad \mathcal{A}_E g_a = -\mathcal{O}_E \mathcal{K}_E \mathcal{O}_E^* g_a, \quad \{\mathcal{O}_E^* g_a : a \in L^2\} = \text{Dom } \mathcal{K}_E.$$

Indeed the endpoint identity first places $\mathcal{O}_E^* g_a$ in the domain; then $i(I - V_E)^{-1}V_E = \mathcal{K}_E/2 - iI/2$ and (2.2) give the formula. Both assertions are on the stated domain of $\mathcal{K}_E$; the first is not an extension of an unbounded composition by density. An eigenvalue $e^{i\theta}$ of U_E , $0 < \theta < 2\pi$, corresponds to $\cot(\theta/2)$ for $\mathcal{K}_E$.

3. ANTIHOLOMORPHIC TRANSMUTATION

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and choose a Riemann map $F : \mathbb{D} \rightarrow \Omega$ with $F(1) = 0$. We use conformal boundary coordinates $\gamma(\theta) = F(e^{i\theta})$. The map F is a smooth diffeomorphism on the closures and F' has no boundary zero [20]. Reparametrization does not change V_E . Let $\mathcal{A}^s$ denote antiholomorphic functions on Ω whose pulled-back traces belong to $H^s(\mathbb{T})$; their traces are exactly the Fourier space with modes $n \leq 0$.

For $h \in \mathcal{A}^s$, $s \geq 1/2$, its antiholomorphic primitive Jh is normalized by $(Jh)(0) = 0$. It is single valued because Ω is simply connected. The identity $(Jh \circ \gamma)' = \overline{\gamma'}(h \circ \gamma)$ implies

$$(3.1) \quad \|Jh \circ \gamma\|_{H^{s+1}} \leq C_s \|h \circ \gamma\|_{H^s}.$$

Evaluation is used only on the primitive, which belongs to H^{s+1} . Define

$$(3.2) \quad \mathcal{V}_E h(z) = \sum_{j=0}^{\infty} \frac{(-E/4)^j}{j!} z^j J^j h(z), \quad \mathcal{J}_E h = |\gamma'| \partial_\nu \mathcal{V}_E h \circ \gamma.$$

The harmonic input in $\mathcal{J}_E$ is identified with its boundary trace. This is the antiholomorphic version of Vekua transmutation [24, 17]. The construction below uses single-valued primitives and (3.1); no star-shapedness assumption is needed.

Lemma 3.1. *The series (3.2) is entire in E in the indicated Sobolev operator norms and satisfies $(-\Delta - E)\mathcal{V}_E h = 0$. The map $\mathcal{V}_E$ is injective. For $s \geq 1/2$, locally uniformly in E ,*

$$(3.3) \quad (\mathcal{V}_E - I) : \mathcal{A}^s \longrightarrow H^{s+1}(\mathbb{T}), \quad (\mathcal{J}_E - |D|) : \mathcal{A}^s \longrightarrow H^s(\mathbb{T})$$

are bounded, with norms $O(E)$ at zero. Here the first map in (3.3) means the boundary trace of $\mathcal{V}_E - I$. In the interior one also has

$$(3.4) \quad \mathcal{V}_E : \mathcal{A}^s \longrightarrow H^{s+1/2}(\Omega), \quad \mathcal{V}_E - I : \mathcal{A}^s \longrightarrow H^{s+3/2}(\Omega).$$

The second map has norm $O(E)$ near zero. The conormal map $\mathcal{J}_E$ is bounded $\mathcal{A}^s \rightarrow H^{s-1}(\mathbb{T})$, with its weak conormal interpretation when $s = 1/2$. For $h \in \mathcal{A}^1$, if $h_0 = h(0)$, then

$$(3.5) \quad \mathcal{O}_E^* \mathcal{J}_E h = i(V_E^* - I)e_0 h_0, \quad (\mathcal{V}_E h) \circ \gamma = h_0 \mathcal{O}_E e_0 - i\mathcal{T}_E \mathcal{J}_E h.$$

Proof. On the trace space H^s , the primitive is bounded and gains one derivative by (3.1). Iterating its H^s bound, and using the gain in the last application, bounds J^j from H^s to H^{s+1} by C_s^j for $j \geq 1$. Multiplication by γ^j has norm at most $C(1+j)^r R^j$ for a fixed integer $r > s+1$. The factorial thus proves convergence, the first gain in (3.3), and parameter continuity. Harmonic extension takes an antiholomorphic trace in $H^r(\mathbb{T})$ into $H^{r+1/2}(\Omega)$. Apply this to h and to each $J^j h$, $j \geq 1$, using one primitive gain before summing. Multiplication by z^j on the fixed smooth domain has a bound of the form $C(1+j)^r R_0^j$ at every fixed Sobolev order. The factorial gives (3.4), including convergence in the interior. Termwise differentiation gives

$$\partial \bar{\partial}(z^j J^j h) = j z^{j-1} J^{j-1} h,$$

which proves the equation. For $j \geq 1$ the conormal trace of a summand is

$$-ij\gamma'\gamma^{j-1}(J^j h) \circ \gamma + i\bar{\gamma}'\gamma^j(J^{j-1} h) \circ \gamma.$$

Its summation proves the conormal claims. Harmonic conormal data in these coordinates are $|D|(h \circ \gamma)$.

For injectivity, termwise differentiation gives $\partial^n \mathcal{V}_E h = (-E/4)^n \mathcal{V}_E J^n h$. The same bounds justify the double series in the identity

$$(3.6) \quad h = \sum_{n=0}^{\infty} \frac{(-z)^n}{n!} \partial^n (\mathcal{V}_E h).$$

For each total degree ℓ , the inner coefficient is $\sum_{n=0}^{\ell} (-1)^n / (n!(\ell-n)!)$, zero unless $\ell = 0$. This proves (3.6), including convergence in boundary H^s .

Antiholomorphic polynomials are dense in $\mathcal{A}^1$ in its trace norm. To see the stated strength of this approximation, first use radial smoothing in conformal coordinates. For an input smooth on the closure, approximate the derivative of its conjugate uniformly by polynomials using Mergelyan's theorem [21]; integrate and fix the value at the boundary origin. Paths in the closure can be chosen of uniformly bounded length. This gives approximation of both boundary values and tangential derivatives, hence in H^1 . For $h = \bar{z}^m$, $m \geq 0$, the series is

$$\sum_{j \geq 0} \frac{(-E/4)^j m!}{j!(m+j)!} z^j \bar{z}^{m+j}.$$

The Herglotz wave with density $e^{-im\phi}$ is $(-ik/2)^m/m!$ times this series. Its initial vector is zero for $m > 0$, and $e_0/\sqrt{2}$ for $m = 0$. The identities (3.5) therefore follow from (2.6)–(2.7) for polynomials, and then for $\mathcal{A}^1$ by the proved bounds. The endpoint value in these identities is only used for H^1 input traces. $\square$

The reverse kernel inclusion uses the following consequence of standard elliptic trace theory [15, 16]. Normal derivatives are weak conormal traces; this distinction matters at the Sobolev index $3/2$.

Lemma 3.2. *Fix $E > 0$ and a smooth domain. An L^2 distributional solution of $(-\Delta - E)w = 0$ has a generalized Dirichlet trace. If that trace belongs to $H^1(\partial\Omega)$, then $w \in H^{3/2}(\Omega)$ and its weak normal derivative belongs to $L^2(\partial\Omega)$. Such solutions satisfy*

$$(3.7) \quad \|w\|_{H^{3/2}(\Omega)} \leq C(\Omega, E)(\|\text{Tr } w\|_{H^1} + \|\partial_\nu^\gamma w\|_{L^2}).$$

Writing $h = \text{Tr } w$ and $g = \partial_\nu^\gamma w$, the traces of the L^2 Helmholtz solutions ∂w and $\bar{\partial} w$ obey, in distributions on the boundary,

$$(3.8) \quad \begin{aligned} h' &= \gamma' \text{Tr}(\partial w) + \bar{\gamma}' \text{Tr}(\bar{\partial} w), \\ g &= -i\gamma' \text{Tr}(\partial w) + i\bar{\gamma}' \text{Tr}(\bar{\partial} w). \end{aligned}$$

Proof. The generalized Dirichlet trace of an L^2 solution is defined by Green's identity in $H^{-1/2}(\partial\Omega)$; see [15, 16]. Let w_b be the harmonic extension of its H^1 trace. Then $w_b \in H^{3/2}$, whereas $w - w_b$ has zero generalized trace and $\Delta(w - w_b) = -Ew \in L^2$. The regularity of the zero-Dirichlet Poisson problem gives $w - w_b \in H^2 \cap H_0^1$. Consequently

$$\|w\|_{H^{3/2}} \leq C(\Omega, E)(\|\text{Tr } w\|_{H^1} + \|w\|_{L^2}),$$

and the normal derivative is in L^2 by the harmonic Dirichlet-to-Neumann map and the H^2 correction. The additional L^2 term can be removed when both Cauchy norms are included: otherwise a sequence normalized in $H^{3/2}$, compact in L^2 , would converge to a nonzero solution with zero Cauchy data. Uniqueness for those data is a contradiction. This proves (3.7), also at eigenvalues. The tangential and normal identities (3.8) follow from Green's identity, first for smooth functions and then for these generalized traces. In particular $\text{Tr}(\partial w) = (h' + ig)/(2\gamma')$. $\square$

We next identify the kernel of the observation adjoint. Lemma 3.1 gives one inclusion; the reverse inclusion requires control of all holomorphic derivatives.

Lemma 3.3. *For every $E > 0$, including a Neumann resonance,*

$$(3.9) \quad \ker \mathcal{O}_E^* = \{\mathcal{J}_E h : h \in \mathcal{A}^1, h(0) = 0\} \quad \text{in } L^2(\mathbb{T}).$$

Proof. One inclusion follows from (3.5). For the converse let $g \in L^2$ with $\mathcal{O}_E^* g = 0$ and solve the driven equation (2.5) with $y(0) = 0$. The variation formula gives $y(L) = 0$, so $y \in H^1(\mathbb{T}; \mathcal{H})$. Set $f_0 = \sqrt{2}y_0$ and $f_n = y_n$ for $n \geq 1$. Pairing with any driven Herglotz vector gives

$$\frac{d}{d\theta} \langle y, y_a \rangle = \frac{i}{2} (\bar{g} h_a - \bar{f}_0 g_a).$$

Both vectors are periodic. Lemma 2.2 therefore gives a Helmholtz solution u with trace f_0 and conormal trace g .

We reconstruct the derivatives without presupposing their regularity. Put $u_0 = u$. The first driven row gives $f'_0 + ig = -ik\gamma' f_1$. By (3.8), $u_1 = (2i/k)\partial u$ has generalized trace $f_1 \in H^1$; Lemma 3.2 therefore makes u_1 an $H^{3/2}$ solution. Inductively, suppose $u_n = (2i/k)^n \partial^n u \in H^{3/2}$ with trace f_n . For $n \geq 1$ the equation gives $\bar{\partial} u_n = -iku_{n-1}/2$. The tangential identity and the n th driven row yield

$$\gamma' \text{Tr}(\partial u_n) = f'_n + \frac{ik}{2} \bar{\gamma}' f_{n-1} = -\frac{ik}{2} \gamma' f_{n+1}.$$

Thus u_{n+1} has generalized trace $f_{n+1} \in H^1$ and is again in $H^{3/2}$. The normal identity gives, for $n \geq 1$,

$$\text{Tr } u_n = f_n, \quad \partial_\nu' u_n = \frac{k}{2} (\bar{\gamma}' f_{n-1} - \gamma' f_{n+1}).$$

The Cauchy estimate is for the same equation at every n , so its constant does not depend on n . Summing its square and using $y \in H^1(\mathbb{T}; \mathcal{H})$ gives

$$(3.10) \quad \sum_{n \geq 0} \|u_n\|_{H^{3/2}(\Omega)}^2 \leq C(\Omega, E)(\|y\|_{H^1(\mathbb{T}; \mathcal{H})}^2 + \|g\|_{L^2}^2) < \infty.$$

In particular the norms of $\partial^n u$ grow at most geometrically.

Since multiplication by z^ℓ on $H^{3/2}(\Omega)$ is bounded by a polynomial in ℓ times R^ℓ , for a fixed R , the factorials below dominate all multiplier and derivative bounds. For $j \geq 0$ define

$$H_j(z) = \sum_{\ell \geq 0} \frac{(-z)^\ell}{\ell!} \partial^{j+\ell} u(z).$$

The series converges in $H^{3/2}(\Omega)$ and in boundary H^1 by (3.10). Differentiation and cancellation show $\partial H_j = 0$ and, for $j \geq 1$, $\bar{\partial} H_j = -(E/4)H_{j-1}$. All $f_n(0)$ vanish, so $H_j(0) = 0$. Consequently

$H_j = (-E/4)^j J^j H_0$ and $H_0 \in \mathcal{A}^1$ has value zero at the origin. The absolutely convergent binomial double series yields

$$u = \sum_{j \geq 0} \frac{z^j}{j!} H_j = \mathcal{V}_E H_0.$$

Taking conormal traces proves the reverse inclusion in (3.9). $\square$

4. THE PERIODIC BOUNDARY FORM

The boundary origin may be chosen separately on a neighborhood of a fixed positive energy. This changes V_E by unitary conjugation and hence does not change its phases or determinant.

Lemma 4.1. *At every $E_0 > 0$ there is a choice of boundary origin and a neighborhood of E_0 on which $c_E = (V_E^* - I)e_0 \neq 0$. Put $s_E = (I + V_E^*)e_0$, $d_E = is_E$ and*

$$(4.1) \quad B_E = \frac{d_E \langle c_E, \cdot \rangle + c_E \langle d_E, \cdot \rangle}{\|c_E\|^2} - \frac{\langle c_E, d_E \rangle}{\|c_E\|^4} c_E \langle c_E, \cdot \rangle, \quad \Pi_E^c = I - \frac{c_E \langle c_E, \cdot \rangle}{\|c_E\|^2}.$$

Here Π_E^c is the orthogonal projection onto $c_E^\perp$. The operator B_E is self-adjoint of rank at most two, $B_E c_E = is_E$, and both families are norm-continuous. At nonresonant energies define

$$(4.2) \quad \mathcal{P}_E = \mathcal{A}_E + \mathcal{O}_E B_E \mathcal{O}_E^*, \quad T_E^\sharp = \Pi_E^c \mathcal{O}_E^*.$$

The first operator extends to a bounded Hermitian map $X \rightarrow X^*$, in the sense that $\langle f, \mathcal{P}_E g \rangle = \overline{\langle g, \mathcal{P}_E f \rangle}$ for $f, g \in X$. The second extends compactly from X to $\mathcal{H}$. Moreover

$$(4.3) \quad \mathcal{P}_E \mathcal{J}_E = 0, \quad T_E^\sharp \mathcal{J}_E = 0 \quad \text{on } \mathcal{A}^{1/2},$$

and, on L^2 at every energy in the neighborhood,

$$(4.4) \quad \ker T_E^\sharp = \text{ran}(\mathcal{J}_E : \mathcal{A}^1 \rightarrow L^2).$$

The map $\mathcal{J}_E : \mathcal{A}^{1/2} \rightarrow X$ is injective there, including at a resonant energy.

Proof. If every rebasing at $\gamma(t)$ fixed e_0 , the original V_{E_0} would fix $W_{E_0}(t)^* e_0$ for every t . These vectors span a dense space by injectivity of $\mathcal{O}_{E_0}$. Thus $V_{E_0} = I$, contradicting (2.9). Choose a rebasing for which $c_{E_0} \neq 0$ and use continuity. Since $\langle c_E, d_E \rangle = -2 \text{Im} \langle e_0, V_E e_0 \rangle \in \mathbb{R}$, (4.1) is self-adjoint and maps c_E to is_E .

For $v \in \mathcal{H}$ the endpoint jump of $\mathcal{O}_E v$ is $\langle c_E, v \rangle$. For $g \in L^2$, the jump of $\mathcal{A}_E g$ is $i \langle s_E, \mathcal{O}_E^* g \rangle$. The added operator in (4.2) cancels this jump because $B_E c_E = is_E$ and the inner product is conjugate-linear in its first variable. The bounded Sobolev extension for $\mathcal{P}_E$ is proved quantitatively in Lemma 4.2 below. Also $\mathcal{O}_E \Pi_E^c : \mathcal{H} \rightarrow H^1(\mathbb{T})$ is Hilbert–Schmidt: it is a bounded smooth interval observation with matching endpoint values, and its first derivative has a rank-one row of uniformly bounded norm. By duality and the compact embedding $H^{-1/2} \hookrightarrow H^{-1}$, $T_E^\sharp : X \rightarrow \mathcal{H}$ is compact and norm-continuous.

Let $p_E = \mathcal{V}_E 1 = J_0(k|z|)$ and $q_E = \mathcal{J}_E 1$, where J_0 denotes the Bessel function of the first kind of order zero. Formula (3.5) and endpoint algebra give

$$(4.5) \quad \mathcal{O}_E^* q_E = ic_E, \quad \mathcal{A}_E q_E = \mathcal{O}_E s_E.$$

The second equality is for nonresonant energies. Thus $\mathcal{P}_E q_E = 0$. For an antiholomorphic h with value zero at the origin, (3.5) gives $\mathcal{O}_E^* \mathcal{J}_E h = 0$ and $\mathcal{A}_E \mathcal{J}_E h = 0$. One can obtain the latter either from polynomial Herglotz approximation and (2.11), or directly from (3.5) and $\mathcal{T}_E + \mathcal{T}_E^* = \mathcal{O}_E \mathcal{O}_E^*$. This proves (4.3) first on $\mathcal{A}^1$, then on $\mathcal{A}^{1/2}$ by density and the proved Sobolev bounds.

If $T_E^\sharp g = 0$ for $g \in L^2$, its observation is a multiple of c_E . Subtract a suitable multiple of q_E , whose observation is ic_E , and apply Lemma 3.3. This proves (4.4), also at a resonance.

Finally suppose $\mathcal{J}_E h = 0$ for $h \in \mathcal{A}^{1/2}$. By (3.4), $u = \mathcal{V}_E h$ belongs to H^1 . It is a weak Neumann eigenfunction, or zero at nonresonance, and is smooth up to the boundary. Since $(\mathcal{V}_E - I)h$ has

$H^{3/2}$ trace, h belongs to $\mathcal{A}^{3/2}$, so its point value is legitimate. The first identity in (3.5) gives $ic_E h(0) = 0$, and hence $h(0) = 0$. The second gives $\text{Tr } u = 0$. Uniqueness for zero Cauchy data makes $u = 0$, and injectivity of $\mathcal{V}_E$ makes $h = 0$. $\square$

We give the regularity details of the periodic correction. They will also permit a bootstrap from the dual trace space back to L^2 .

Lemma 4.2. *At a nonresonant energy the normalized self-adjoint operator*

$$(4.6) \quad \hat{P}_E = \Lambda^{1/2} \mathcal{P}_E \Lambda^{1/2} = 4\Pi_+ + C_E$$

is bounded on L^2 , where C_E is compact. Locally its compact part is norm-continuous. For any fixed $0 < \delta < 1/2$, the compact remainder C_E , not $\hat{P}_E$, is bounded $H^t \rightarrow H^{t+\delta}$ for $0 \leq t \leq 1/2$, locally uniformly. If $E_0 = \mu$ is a Neumann eigenvalue, then

$$(4.7) \quad \hat{P}_E = -\frac{2}{E - \mu} \sum_{j=1}^m (\Lambda^{1/2} h_j) \langle \Lambda^{1/2} h_j, \cdot \rangle + \hat{P}_{\text{reg}}(E), \quad \hat{P}_{\text{reg}}(E) = 4\Pi_+ + C_{\text{reg}}(E),$$

where $C_{\text{reg}}(E)$ has the same continuity and regularizing properties across μ . Moreover

$$(4.8) \quad \hat{J}_E := \Lambda^{-1/2} \mathcal{J}_E \Lambda^{-1/2} : \mathcal{H}_{\leq 0} \longrightarrow L^2$$

is the inclusion plus a compact operator gaining one derivative.

Proof. On mean-zero conormal data the harmonic Neumann map, up to its additive constant, is $|D|^{-1}$. Choose the harmonic solution u_0 with zero volume mean. Then

$$(4.9) \quad u_E = u_0 + E(A_N - E)^{-1} u_0,$$

where A_N is the Neumann Laplacian. The correction maps $H^{-1/2}$ boundary data into $H^3(\Omega)$, and hence its trace into $H^{5/2}$. Constants contribute smooth finite-rank operators. Resolvent expansion at μ , or the same formula using a fixed nonresonant reference energy for arbitrary data, gives

$$(4.10) \quad \mathcal{N}(E) = -\frac{1}{E - \mu} \sum_{j=1}^m h_j \langle h_j, \cdot \rangle + \mathcal{N}_{\text{reg}}(E).$$

For a direct normalization check, let $P_{-1}g$ solve the Neumann problem at energy -1 . Then $u_E = P_{-1}g + (E + 1)(A_N - E)^{-1}P_{-1}g$ and $\langle \phi_j, P_{-1}g \rangle = \langle h_j, g \rangle / (\mu + 1)$. This proves (4.10). Its regular remainder has the harmonic principal part just described and the additional smoothness used here.

The periodic integral kernel of $2D^{-1}$, defined to vanish on constants, is $i \operatorname{sgn}(\theta - s) - i(\theta - s)/\pi$ on the cut square $[0, L]^2$. Subtracting it from $i(\mathcal{T}_E - \mathcal{T}_E^*) + \mathcal{O}_E B_E \mathcal{O}_E^*$ leaves the kernel

$$(4.11) \quad \begin{aligned} r_E(\theta, s) &= i \operatorname{sgn}(\theta - s)(w_E(\theta, s) - 1) + i(\theta - s)/\pi \\ &+ \langle e_0, W_E(\theta) B_E W_E(s)^* e_0 \rangle, \quad w_E(\theta, s) = \langle e_0, W_E(\theta) W_E(s)^* e_0 \rangle. \end{aligned}$$

Here $w_E(s, s) = 1$ and $\partial_\theta w_E(s, s) = 0$. Thus the first term is C^1 across the diagonal. The endpoint cancellation in Lemma 4.1 shows that opposite-edge values of (4.11) match. The kernel is smooth on the pieces of the cut square and has matching values, so it is in $H^{1+\delta}(\mathbb{T}^2)$ for every $\delta < 1/2$. One way to check this last assertion is to differentiate once: the resulting bounded piecewise smooth functions, with jumps only across finitely many smooth lines, belong to H^δ for $\delta < 1/2$ by the fractional difference-quotient norm. First derivatives need not match at the cut. All these estimates are locally uniform and continuous.

If r_{mn} are its Fourier matrix entries, then for $0 \leq t \leq 1/2$

$$\sum_{m,n} (1 + |m|)^{1+2t+2\delta} (1 + |n|)^{1-2t} |r_{mn}|^2 \leq C_\delta \|r_E\|_{H^{1+\delta}(\mathbb{T}^2)}^2.$$

Thus sandwiching the remainder by $\Lambda^{1/2}$ gives a compact operator $H^t \rightarrow H^{t+\delta}$. The combined principal part is $2|D|^{-1} + 2D^{-1} = 4D_+^{-1}\Pi_+$; its normalization is $4\Pi_+$ plus an operator gaining one derivative. The Neumann comparison above proves (4.6) and (4.7). At higher Sobolev orders in the stated range, the same comparison uses the corresponding smooth-boundary elliptic estimates. The pole-subtracted resolvent and the kernels depend continuously on E in these norms. This also proves the $X \rightarrow X^*$ extension asserted in Lemma 4.1.

Finally the principal part of (4.8) is $\Lambda^{-1/2}|D|\Lambda^{-1/2}$ on nonpositive modes. It differs from the inclusion by an operator gaining one derivative, including a one-dimensional discrepancy on constants. Equation (3.3) gives the same gain for its remainder. $\square$

5. UNIFORM BOUNDS FOR THE CAYLEY TRANSFORM

We now fix any positive E_0 , choose the local cut from Lemma 4.1, and work on its energy neighborhood. All neighborhoods below may be decreased. They contain no other Neumann eigenvalue. The symbol m is zero at a nonresonant E_0 , and otherwise its Neumann multiplicity. When $m = 0$, regular parts mean the original operators, $X_0 = X$, and $Z = \mathcal{H}_+$. Empty sums and compatibility conditions are then omitted.

Lemma 5.1. *There is a fixed map $R : \mathcal{H}_+ \rightarrow L^2$, equal to the inclusion plus a finite-rank map with finite Fourier input and output vectors, such that*

$$(5.1) \quad \mathfrak{B}_E(a, z) = \hat{J}_E a + Rz : \mathcal{H}_{\leq 0} \oplus \mathcal{H}_+ \longrightarrow L^2$$

is invertible locally in E , with norm-continuous inverse. Put

$$\mathcal{R}_E = R^* \hat{P}_E R, \quad \mathcal{S}_E = T_E^\# \Lambda^{1/2} R.$$

After its pole is subtracted, the first operator is $4I$ plus a compact remainder having the regularizing properties of Lemma 4.2. The second operator is compact and norm-continuous. At a resonance put $\eta_j = R^ \Lambda^{1/2} h_j$ and*

$$Z = \{z \in \mathcal{H}_+ : \langle \eta_j, z \rangle = 0 \ (1 \leq j \leq m)\}.$$

The space Z has codimension m , its orthogonal complement consists of smooth functions, and the pole of $\mathcal{R}_E$ vanishes on Z . We write Π_Z for the orthogonal projection onto Z .

Proof. The map $\hat{J}_{E_0} \Pi_{\leq 0} + \Pi_+$ is $I +$ compact, hence Fredholm of index zero. Its restriction to $\mathcal{H}_{\leq 0}$ is injective by Lemma 4.1. Consequently projection of its kernel onto the positive input space is injective. A finite-rank map acting only on positive inputs can therefore pair this kernel isomorphically with the cokernel and make the whole operator invertible. Approximate its finitely many input and output vectors by finite Fourier sums; invertibility is open, so the correction can have the stated smooth vectors. Fix it as $R - \Pi_+$. Continuity proves (5.1) locally. The reduced operator assertions follow from Lemma 4.2.

Every $\mathcal{J}_{E_0} h$ is the normal trace of a Helmholtz solution at E_0 , and hence is orthogonal to each h_j . Thus $\Lambda^{1/2} h_j$ annihilates $\text{ran } \hat{J}_{E_0}$. If a combination of the η_j is zero, the corresponding combination of $\Lambda^{1/2} h_j$ also annihilates $\text{ran } R$, and hence all of L^2 by (5.1). The independence in Lemma 2.2 gives the claim. Formula (4.7) now gives

$$(5.2) \quad \mathcal{R}_E = -\frac{2}{E - E_0} \sum_{j=1}^m \eta_j \langle \eta_j, \cdot \rangle + \mathcal{R}_{\text{reg}}(E), \quad \mathcal{R}_{\text{reg}}(E) = 4I + \text{compact}.$$

The fact that R is fixed, rather than energy dependent, makes Z fixed and cancels the pole there exactly. $\square$

Lemma 5.2. *The restriction $\mathcal{S}_{E_0}|_Z$ is injective. At a nonresonant energy this means injectivity on all of $\mathcal{H}_+$.*

Proof. Let $z \in Z$ satisfy $\mathcal{S}_{E_0}z = 0$ and set $g = \Lambda^{1/2}Rz \in X$. Initially this is only a dual trace, not an L^2 function. It belongs to X_0 from (2.10).

For Herglotz data at E_0 , endpoint identities without an inverse give

$$(5.3) \quad \mathcal{P}_{\text{reg}}(E_0)g_a = \mathcal{O}_{E_0}\xi_a - 2 \sum_{j=1}^m h_j \langle \phi_j, u_a \rangle, \quad \xi_a = \sqrt{2}[(I + V^*) + iB(V^* - I)]v_a \in c^\perp.$$

Here $\mathcal{P}_{\text{reg}}$ has the Neumann resolvent pole removed, and all unmarked operators on the right are at E_0 . At nonresonance it is simply $\mathcal{P}$. To verify this, use $2h_a + i(\mathcal{T} - \mathcal{T}^*)g_a = \sqrt{2}\mathcal{O}(I + V^*)v_a$ and $\mathcal{O}^*g_a = \sqrt{2}i(V^* - I)v_a$. The identity $Bc = is$ shows directly that $\langle c, \xi_a \rangle = 0$. At resonance $\mathcal{N}_{\text{reg}}(E_0)g_a$ is the trace of the solution orthogonal to the eigenspace, namely $h_a - \sum h_j \langle \phi_j, u_a \rangle$, which proves (5.3).

Pair (5.3) with g . The finite sum vanishes by compatibility. The observation term vanishes because $T_{E_0}^\sharp g = 0$ and $\xi_a \in c^\perp$. These pairings are legitimate on X : $\mathcal{O}\xi_a$ is periodic H^1 , whereas the raw observation need not be periodic. Density in Lemma 2.2 implies

$$\langle g, \mathcal{P}_{\text{reg}}(E_0)\tilde{g} \rangle = 0 \quad (\tilde{g} \in X_0).$$

In particular, with $\tilde{g} = \Lambda^{1/2}Rw$, $w \in Z$, this says

$$(5.4) \quad \Pi_Z \mathcal{R}_{\text{reg}}(E_0)z = 0, \quad 4z = -\Pi_Z C z, \quad C = \mathcal{R}_{\text{reg}}(E_0) - 4I.$$

The projection Π_Z differs from the identity by a smooth finite-rank map. Take $\delta = 1/4$ in Lemma 4.2; twice applying (5.4) gives $z \in H^{1/2}$, so $g \in L^2$.

We may therefore apply (4.4). It gives $g = \mathcal{J}_{E_0}h$ for $h \in \mathcal{A}^1$. After normalization this lies in $\text{ran } \hat{J}_{E_0}$. But it is also Rz . Uniqueness in the decomposition (5.1) forces $z = 0$. This proves injectivity using the kernel identity only at its proved L^2 regularity. $\square$

Lemma 5.3. *Let $\mathcal{Z}$ and $\mathcal{Y}$ be Hilbert spaces. Let $H = \alpha I + C$ be self-adjoint on $\mathcal{Z}$, with $\alpha > 0$ and C compact, and let $S : \mathcal{Z} \rightarrow \mathcal{Y}$ be bounded and injective. There are constants $A > 0$ and $\varepsilon > 0$ such that*

$$H + AS^*S \geq \varepsilon I.$$

Proof. If no integer n gives a strictly positive lower bound for $H + nS^*S$, choose unit vectors z_n with $\langle z_n, H z_n \rangle + n\|S z_n\|^2 \leq 1/n$. The first term is bounded, so $S z_n \rightarrow 0$. Pass to a weakly convergent subsequence. Its limit lies in $\ker S$ and is zero. Compactness gives $\langle z_n, C z_n \rangle \rightarrow 0$, so $\langle z_n, H z_n \rangle \rightarrow \alpha$, a contradiction. The resulting inequality remains valid with $\varepsilon/2$ in place of ε under sufficiently small operator-norm perturbations of H and S , with the same A . $\square$

Proposition 5.4. *If $E_0 > 0$ is nonresonant, $\mathcal{K}_E \leq AI$ uniformly on a neighborhood of E_0 . If $E_0 = \mu > 0$ has Neumann multiplicity m , there are $A > 0$ and $a > 0$ such that*

$$(5.5) \quad \text{rank } \mathbf{1}_{(A, \infty)}(\mathcal{K}_E) \leq m \quad (0 < |E - \mu| < a).$$

Only nonresonant energies are used in this formula.

Proof. Apply Lemma 5.3 on the fixed space Z to $\Pi_Z \mathcal{R}_{\text{reg}}(E_0)|_Z$ and $\mathcal{S}_{E_0}|_Z$. The hypotheses follow from Lemmas 5.1 and 5.2. By continuity and (5.2), locally and off resonance,

$$(5.6) \quad \langle z, \mathcal{R}_E z \rangle \geq -A_0 \|\mathcal{S}_E z\|^2 \quad (z \in Z).$$

For any $g \in X$ write $\Lambda^{-1/2}g = \hat{J}_E a + Rz$ using (5.1). Equations (4.3) imply

$$\langle g, \mathcal{P}_E g \rangle = \langle z, \mathcal{R}_E z \rangle, \quad T_E^\sharp g = \mathcal{S}_E z.$$

Impose the at most m linear conditions $z \in Z$ on g . For $g \in L^2$ obeying them, (5.6) and boundedness of B_E give

$$(5.7) \quad \langle g, \mathcal{A}_E g \rangle \geq -(A_0 + \|B_E\|)\|\mathcal{O}_E^* g\|^2.$$

The norm of B_E is bounded on the chosen neighborhood.

Suppose $\mathcal{K}_E$ had an $(m+1)$ -dimensional spectral subspace strictly above a larger uniform constant A . Lift a basis to Herglotz normal traces by the range identity in (2.11). Their span maps isomorphically to that subspace. At least one nonzero linear combination obeys the m linear conditions, and (2.11) makes its quadratic form strictly less than $-A\|\mathcal{O}_E^*g\|^2$, contradicting (5.7). This proves (5.5). For $m=0$ the argument gives the full upper bound. In particular it gives a uniform spectral gap immediately above 1 for U_E off the Neumann spectrum. It does not assume that every phase arriving at 1 has a nonzero limiting eigenvector. $\square$

6. SMALL-ENERGY NORMALIZATION

The counting argument requires a normalization at zero. The following rank-one decomposition shows that the zero Neumann mode contributes exactly one. For a unitary sufficiently close to the identity, a principal eigenphase means an argument in $(-\pi, \pi)$.

Lemma 6.1. *For every smooth Jordan domain and all sufficiently small $E > 0$,*

$$(6.1) \quad \frac{E|\Omega|/2 - \Phi_-(E)}{2\pi} = 1.$$

Proof. Restrict the energy interval to $(0, \mu_1(\Omega))$ and keep a fixed conformal parametrization and base point. With $p_E = \mathcal{V}_E 1$, $q_E = \mathcal{J}_E 1$, set $b_E = \mathcal{A}_E q_E$ and $a_E = \langle q_E, b_E \rangle$. Equations (4.5), (2.4) and (2.3) give

$$b_E = \mathcal{O}_E(I + V_E^*)e_0 = 2 + \beta_E, \quad \|\beta_E\|_{C^1[0,L]} = O(E), \quad a_E = 2\operatorname{Im}\langle e_0, V_E e_0 \rangle = -2E|\Omega| + o(E) < 0.$$

The zeroth-row matrix elements of the transport have no linear term in k , which justifies the C^1 estimate. The jump of $\mathcal{A}_E g$ is $i\langle b_E, g \rangle$, so the self-adjoint Schur complement

$$(6.2) \quad C_E^{\text{sc}} = \mathcal{A}_E - a_E^{-1} b_E \langle b_E, \cdot \rangle$$

has matching endpoint values and annihilates q_E and every $\mathcal{J}_E h$ with $h(0) = 0$.

On $\mathcal{H}_+$ put $S_+ = D_+^{-1/2}$. Its positive Fourier block is

$$(6.3) \quad \langle g, C_E^{\text{sc}} g \rangle = \langle S_+ g, (4I + K_E^{\text{sc}}) S_+ g \rangle, \quad \|K_E^{\text{sc}}\| = O(E).$$

Here is an explicit check of the cut term in this assertion. The remainder relative to $i(\mathcal{T}_0 - \mathcal{T}_0^*)$, where $\mathcal{T}_0 g(\theta) = \int_0^\theta g$, is $r_E^0 = i \operatorname{sgn}(\theta - s)(w_E - 1)$, a C^1 kernel of size $O(E)$ on the cut square. Direct endpoint evaluation gives

$$r_E^0(L, s) - r_E^0(0, s) = i\overline{\beta_E(s)}, \quad \beta_E(L) - \beta_E(0) = ia_E.$$

Thus $r_E^0 - \beta_E(\theta)\overline{\beta_E(s)}/a_E$ is a periodic Hermitian $H^1(\mathbb{T}^2)$ kernel with norm $O(E)$. Its normalization by $D_+^{1/2}$ on both sides is Hilbert–Schmidt, because $mn \leq (m^2 + n^2)/2$. On mean-zero data the compression of the rank-one term using b_E equals the one using β_E . Formula (4.9), with zero volume mean, has norm $O(E)$ at zero after its smoothing gain, and direct Fourier integration gives $\langle g, i(\mathcal{T}_0 - \mathcal{T}_0^*)g \rangle = 2 \sum_{n>0} |g_n|^2/n$. The harmonic Neumann term contributes the other $2 \sum |g_n|^2/n$. These observations prove (6.3).

For $g \in \mathcal{H}_-$ define $h_g \in \mathcal{A}^1$ by

$$h_g \circ \gamma = \sum_{n<0} \frac{g_n}{|n|} (\varphi_n - \varphi_n(0)).$$

It has value zero at the origin and harmonic conormal density g . Equation (3.3) gives $G_E g := \mathcal{J}_E h_g = g + O(E)g$ in L^2 . Also

$$\int_0^L q_E d\theta = -E \int_\Omega p_E = -E|\Omega| + O(E^2), \quad q_E/E \longrightarrow -\frac{1}{2} \operatorname{Im}(\bar{\gamma}\gamma') \quad \text{in } L^2.$$

The vector $\zeta_E = q_E / \langle \varphi_0, q_E \rangle$ therefore has constant Fourier coefficient one and converges in L^2 . The map

$$\tilde{B}_E = \Pi_+ + G_E \Pi_- + \zeta_E \langle \varphi_0, \cdot \rangle$$

is invertible for small E : its limit is I plus the square-zero rank-one map $(\zeta_0 - \varphi_0) \langle \varphi_0, \cdot \rangle$. The operator C_E^{sc} annihilates $G_E \mathcal{H}_-$ and ζ_E . These inclusions and the invertibility of $\tilde{B}_E$, together with (6.3), give $C_E^{\text{sc}} \geq 0$ on all of L^2 . No converse nullspace assertion is needed here. Consequently $\text{ind}_- \mathcal{A}_E = 1$: it is at most one by (6.2), and at least one since $a_E < 0$.

By (2.11), every positive Cayley spectral subspace lifts to a negative subspace for $\mathcal{A}_E$, so there is at most one positive principal phase when $\|U_E - I\| < 1$. The principal logarithm has trace $iE|\Omega|/2$ by (2.3) and continuity down to zero. This positive sum shows that at least one positive principal phase exists. Passing to the negative argument branch subtracts 2π exactly once. This proves (6.1). $\square$

7. THE COUNTING INEQUALITY

Let $\text{Arg}_- z \in (-2\pi, 0)$ denote the negative argument of a point z on the unit circle other than 1. At any positive nonresonant E , Proposition 5.4 with $m = 0$ shows that $\mathcal{K}_E$ is bounded above. Its compact resolvent implies that it has only finitely many positive eigenvalues. Hence U_E has only finitely many upper-semicircle eigenvalues, while $|\text{Arg}_- z| \leq C|z - 1|$ on the lower semicircle. The series

$$\Phi_-(E) = \sum_{z \in \sigma_p(U_E) \setminus \{1\}} \text{Arg}_- z$$

is therefore absolutely convergent, with multiplicities. The determinant formula makes (1.2) an integer. For $|z| = 1$, $z \neq 1$,

$$-\text{Arg}_- z \geq |z - 1|.$$

The compact normal operator $U_E - I$ has its norm attained at a nonzero spectral value whenever it is nonzero. Therefore

$$(7.1) \quad -\Phi_-(E) \geq \|U_E - I\|, \quad M(E) \geq \frac{E|\Omega|}{4\pi} + \frac{\|U_E - I\|}{2\pi}.$$

On a compact nonresonant interval, the uniform upper Cayley bound gives a uniform empty arc immediately above 1. Put a logarithm cut in that arc and normalize the logarithm to be zero at 1. For this logarithm, $\log z = (z - 1)f(z)$ with f holomorphic near the spectrum. Functional calculus and trace-norm continuity of $U_E - I$ therefore make $\log U_E$ trace-norm continuous. Its trace is $i\Phi_-(E)$. Thus M is constant on each nonresonant interval.

At a positive Neumann eigenvalue μ of multiplicity m , take A, a from (5.5). Choose a sufficiently small $\varepsilon > 0$ with $\cot(\varepsilon/2) > A$ and $e^{i\varepsilon} \notin \sigma(U_\mu)$. This is possible since all spectral points other than 1 are isolated finite-multiplicity eigenvalues. By operator-norm continuity, the cut through $e^{i\varepsilon}$ stays off the spectrum on a smaller neighborhood. Let $\log_\varepsilon$ be the holomorphic branch on the plane cut through $e^{i\varepsilon}$ with $\log_\varepsilon(1) = 0$, and put $\arg_\varepsilon U_E = -i \log_\varepsilon U_E$. Its values on the unit circle have arguments in $(\varepsilon - 2\pi, \varepsilon)$. Its trace is continuous through μ , even on actual fixed vectors. Therefore

$$J_\varepsilon(E) = \frac{E|\Omega|/2 - \text{tr } \arg_\varepsilon U_E}{2\pi}$$

is an integer constant through μ . Write $n_{(0,\varepsilon)}(U_E)$ for the number of eigenvalues of U_E in the arc $\{e^{it} : 0 < t < \varepsilon\}$, counted with multiplicity. At nonresonant nearby energies,

$$(7.2) \quad M(E) = J_\varepsilon(E) + n_{(0,\varepsilon)}(U_E), \quad 0 \leq n_{(0,\varepsilon)}(U_E) \leq m.$$

The second inequality is exactly (5.5). Thus the net increase of M across μ is at most m . Notice that no eigenvector continuation at 1 is used in this argument.

Lemma 6.1 gives $M = 1$ below the first positive Neumann eigenvalue. There are finitely many resonances below any fixed energy. Summing (7.2) gives, for every nonresonant $E > 0$,

$$M(E) \leq 1 + \sum_{0 < \mu < E} \dim \ker(A_N - \mu) = N_N(E).$$

Together with (7.1), this yields

$$(7.3) \quad N_N(E) \geq \frac{E|\Omega|}{4\pi} + \frac{\|U_E - I\|}{2\pi}.$$

For a resonant energy $\mu > 0$, approach from below through nonresonant energies. The strict counting function is constant just below μ and equals $N_N(\mu)$ there. Norm continuity of U_E therefore proves (7.3) at μ as well. Since $N_N(\mu_j) \leq j$, we have proved the quantitative eigenvalue bound

$$(7.4) \quad |\Omega|\mu_j + 2\|U_{\mu_j} - I\| \leq 4\pi j \quad (j \geq 1).$$

The fixed-space estimate (2.9) rules out $U_E = I$ for every $E > 0$: the identity would have an infinite-dimensional fixed space. Thus the inequalities in Theorem 1 are strict for smooth domains. The quantitative form (7.4) is needed for the nonsmooth limit.

8. LIPSCHITZ DOMAINS

For a Lipschitz Jordan domain, choose a positively oriented rescaled arclength parametrization $\gamma : [0, L] \rightarrow \partial\Omega$, based at zero. Equation (2.1) has a unitary, absolutely continuous solution, since its coefficient is in L^∞ . Orientation-preserving bi-Lipschitz reparametrizations do not change V_E . Changing the base point conjugates V_E by a unitary transport matrix. In particular $\|U_E - I\|$ is independent of these choices.

Lemma 8.1. *For every bounded simply connected Lipschitz domain and every $E > 0$,*

$$(8.1) \quad \dim \ker(V_E - I) \leq \dim \ker(-\Delta_\Omega^N - E) < \infty.$$

Proof. The observation $\mathcal{O}_E$ is injective also for this parametrization. Indeed, if $f = W_E v$ has $f_0 = 0$, the first transport row and $|\gamma'| > 0$ almost everywhere imply $f_1 = 0$ almost everywhere; absolute continuity and the successive rows give $f_n = 0$ for every n .

If $V_E v = v$, then $f = W_E v$ is periodic and $f_0 \in W^{1,\infty}(\mathbb{T})$. The Herglotz driven identities hold almost everywhere on the Lipschitz boundary. Pairing f with a driven vector and integrating gives $\langle f_0, g_a \rangle = 0$ for every Herglotz density a . The outgoing double-layer potential with density f_0 has zero far field, by conjugation and reversal of plane-wave directions. Rellich uniqueness and the connectedness of the exterior make it zero outside. The negative of this double-layer potential is an interior H^1 solution with zero weak Neumann data and Dirichlet trace f_0 , by the Lipschitz mapping and jump relations in [16]. The trace identifies this map injectively, by injectivity of $\mathcal{O}_E$. The compact embedding $H^1(\Omega) \hookrightarrow L^2(\Omega)$ makes the Neumann eigenspace finite-dimensional, proving (8.1). Since $\mathcal{H}$ is infinite-dimensional, this also proves $U_E \neq I$. $\square$

Lemma 8.2. *Let γ_n, γ be closed Lipschitz curves on $[0, L]$, all based at zero, with $\gamma_n \rightarrow \gamma$ uniformly and $\sup_n \int_0^L |\gamma'_n| < \infty$. If $E_n \rightarrow E > 0$, then their transport monodromies satisfy*

$$(8.2) \quad \|U_{\gamma_n}(E_n) - U_\gamma(E)\| \rightarrow 0.$$

Proof. Write $\gamma_n = x_n + iy_n$, $k_n = \sqrt{E_n}$, and use the bounded skew-adjoint operators G_x, G_y from Lemma 2.1. Set

$$C_n = k_n(x'_n G_x + y'_n G_y), \quad C = k(x' G_x + y' G_y), \quad F_n = k(x G_x + y G_y) - k_n(x_n G_x + y_n G_y).$$

Then $F'_n = C - C_n$, $F_n(0) = F_n(L) = 0$, and $\|F_n\|_\infty \rightarrow 0$. If $W'_n = C_n W_n$ and $W' = CW$, integration by parts in the absolutely continuous operator-valued functions gives

$$\begin{aligned} V_n^* V - I &= \int_0^L W_n^* F'_n W \, d\theta \\ &= \int_0^L W_n^* (C_n F_n - F_n C) W \, d\theta. \end{aligned}$$

Unitarity yields

$$\|V_n^* V - I\| \leq \|F_n\|_\infty \int_0^L (\|C_n\| + \|C\|) \, d\theta \rightarrow 0.$$

This proves (8.2). In particular, convergence of the derivatives γ'_n is not required. $\square$

We now apply smooth approximation. Neumann domain monotonicity is neither assumed nor used. Theorem 5.1 and Remark 5.3 of Ball–Zarnescu [1] provide smooth domains Ω_ϵ and uniformly bi-Lipschitz maps $f_\epsilon : \overline{\Omega} \rightarrow \overline{\Omega_\epsilon}$, preserving topology and equal to the identity outside a boundary collar shrinking to zero. In particular Ω_ϵ is simply connected. Here $\epsilon \downarrow 0$ may be chosen for the inner approximation. The same construction is jointly continuous at $\epsilon = 0$, so $\sup_{\overline{\Omega}} |f_\epsilon - \text{id}| \rightarrow 0$. Pulling the Neumann form back to Ω gives

$$q_\epsilon[u] = \int_\Omega A_\epsilon \nabla u \cdot \nabla \bar{u}, \quad m_\epsilon[u] = \int_\Omega \rho_\epsilon |u|^2,$$

where $\rho_\epsilon = |\det Df_\epsilon|$ and $A_\epsilon = \rho_\epsilon Df_\epsilon^{-1} Df_\epsilon^{-T}$. They are uniformly bounded and uniformly positive; they equal 1 and I , respectively, off the shrinking collar. For each fixed finite-dimensional subspace of $H^1(\Omega)$, dominated convergence of these forms and min–max give $\limsup \mu_j(\Omega_\epsilon) \leq \mu_j(\Omega)$. Conversely, pull back the first $j + 1$ orthonormal eigenfunctions. They are bounded in $H^1(\Omega)$ and have a subsequence converging weakly in H^1 and strongly in L^2 . Weighted orthonormality passes to ordinary orthonormality, since ρ_ϵ is bounded and tends to one almost everywhere. On each fixed interior subdomain the energy is eventually the ordinary Dirichlet energy, so weak lower semicontinuity followed by exhaustion gives the required lower energy bound for every linear combination of the limits. More explicitly, after taking a subsequence realizing the lower limit, the limiting functions $u_0, \dots, u_j$ are orthonormal and satisfy

$$q \left[\sum_{\ell=0}^j a_\ell u_\ell \right] \leq \left(\liminf_{\epsilon \downarrow 0} \mu_j(\Omega_\epsilon) \right) \sum_{\ell=0}^j |a_\ell|^2.$$

The min–max principle gives $\mu_j(\Omega) \leq \liminf \mu_j(\Omega_\epsilon)$. Thus

$$\mu_j(\Omega_\epsilon) \rightarrow \mu_j(\Omega), \quad |\Omega_\epsilon| = \int_\Omega \rho_\epsilon \rightarrow |\Omega|.$$

For the boundary transport use

$$\gamma_\epsilon = f_\epsilon \circ \gamma - f_\epsilon(\gamma(0)).$$

These curves parametrize translated smooth boundaries with positive orientation. Their lengths are uniformly bounded, by the uniform Lipschitz bound on f_ϵ , and they converge uniformly to γ . Although these particular parametrizations need not be smooth, reparametrization invariance identifies their monodromies with the ones used in the smooth-domain proof. Lemma 8.2 and eigenvalue convergence imply

$$U_{\gamma_\epsilon}(\mu_j(\Omega_\epsilon)) \rightarrow U_\gamma(\mu_j(\Omega)) \quad \text{in operator norm.}$$

Passing (7.4), rather than only its strict consequence, to the limit gives

$$(8.3) \quad |\Omega| \mu_j(\Omega) + 2 \|U_\gamma(\mu_j(\Omega)) - I\| \leq 4\pi j \quad (j \geq 1).$$

The additional term in (8.3) is positive by Lemma 8.1. Thus $|\Omega|\mu_j(\Omega) < 4\pi j$. If $m = N_N(E)$, then $E \leq \mu_m(\Omega) < 4\pi m/|\Omega|$, proving the strict counting inequality as well. This completes the proof of Theorem 1.

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